

The Brownian Matrix Governing Fabius Smoothness

Sharp anisotropic Fourier envelopes,
mixed derivative growth, and a signed resonance

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Abstract

The common-digit geometric-uniform construction couples several Fabius-type laws using one sequence of independent uniform digits. Its smooth joint density is known in the ProveIt corpus, but sharp joint Fourier decay is left conjectural. We prove a uniform logarithmic-box envelope for the joint infinite sinc product, including matching lower bounds on a fixed positive proportion of every box. The controlling function is the area under the upper envelope of finitely many lines. Its positive-orthant convex conjugate is exactly the quadratic form associated with the matrix $M_{ij} = \min(\lambda_i, \lambda_j)$, where $\lambda_j = -\log |q_j|$. Consequently, for every mixed derivative and every $1 \leq p \leq \infty$,

$$\log \|\partial^\alpha f\|_{L^p} = \frac{1}{2} \alpha^\top M \alpha + O(|\alpha| + 1),$$

with an error uniform over all multi-indices and all such p . This identifies the complete quadratic-order anisotropic smoothness law, not just an isotropic upper estimate. We also prove sharp ray-window decay, a reciprocal convolution-power law, directional derivative stratification, and quantitative boundary flatness. Distinct parameter moduli are essential: at the signed pair $(r, -r)$ the leading derivative-growth coefficient doubles. The resulting discontinuity yields two noncommuting parameter/order limits. All principal arguments are given in full. Exact finite algebra checks and high-precision diagnostics accompany the article; the analytic proofs have not been checked by a proof assistant.

Status. Unrefereed research draft. The precise derivative law and its multivariate envelope proof are contributions developed here relative to the inspected source. No claim of exhaustive historical priority is made.

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1 The research gap and the principal results

1.1 What is already established, and what is new here

For independent $U_k \sim \text{Uniform}[0, 1]$, the normalized variables

$$X_q = (1 - q) \sum_{k=0}^{\infty} q^k U_k, \quad 0 < q < 1, \quad (1.1)$$

provide the common-digit geometric-uniform family: the same U_k is used at each parameter. The one-dimensional $q = 1/2$ law belongs to the classical Fabius–Rvachev theory [2, 3]. The repository report *Common-Digit Fabius Zonoids* already proves its multivariate support formula, a joint infinite sinc product, C_c^∞ regularity, boundary flatness, and exact support volumes [1]. Those facts are not new claims of the present article.

The explicit unresolved item is the report’s `conj : ray-decay`, in the subsection “Sharp ray-wise decay of the joint sinc product.” It predicts a quadratic logarithmic leading term determined by the largest surviving parameter, while also proposing a more delicate periodic/quasiperiodic description of the remainder. Its reference to neighborhoods of the zero hyperplanes does not prescribe a precise exceptional set. Our answer separates these issues. We prove the sharp leading envelope with a precise window and measure formulation, then strengthen it to frequency directions that move with scale. We do *not* prove the proposed periodic/quasiperiodic remainder description.

The main additional result is the exact mixed-derivative growth law. It contains cross terms that are absent for independent marginals. These terms are the analytic signature of sharing the same digits.

The source was inspected at commit 23dd71d2d3d5e85d2b97a8cc45f55e735d69a4ae. The relevant source labels are `thm: joint-char`, `thm: density-regularity`, and `conj: ray-decay`. The package’s `PROVENANCE.md` records the exact path and the inspection boundary. A targeted search did not locate the mixed-derivative formula below in the inspected report or the retrieved related material. That is a bounded source comparison, not a claim to have searched all mathematical literature.

1.2 A normalization that exposes the Fourier geometry

Let $V_k = 2U_k - 1$, so V_k is uniform on $[-1, 1]$. We use the centered variables

$$Y_j = \sum_{k=0}^{\infty} q_j^k V_k, \quad 1 \leq j \leq d. \quad (1.2)$$

Unless explicitly stated otherwise, assume

$$0 < |q_j| < 1, \quad |q_i| \neq |q_j| \quad (i \neq j), \quad \lambda_j = -\log |q_j|. \quad (1.3)$$

Thus signed parameters are allowed, but repeated moduli are not. Let f be the density of $Y = (Y_1, \dots, Y_d)$. We prove its existence and smoothness below. For the positive normalized family,

$$X_{q_j} = \frac{1}{2} + \frac{1 - q_j}{2} Y_j. \quad (1.4)$$

Fixed nonzero diagonal changes of coordinates alter logarithms of derivative norms by only $O(|\alpha| + 1)$. Hence the leading results are identical for (1.1) and (1.2).

Write

$$M_{ij} = \min(\lambda_i, \lambda_j), \quad \mathcal{Q}(\alpha) = \frac{1}{2} \alpha^\top M \alpha, \quad |\alpha| = \sum_j \alpha_j. \quad (1.5)$$

All logarithms are natural. Multi-indices belong to $\mathbb{N}^d$, with zero included in $\mathbb{N}$. Constants may depend on the fixed parameter vector and on d ; they are not asserted to remain bounded as moduli collide.

Theorem 1.1 (Sharp mixed-derivative law). *Under (1.3), $f \in C_c^\infty(\mathbb{R}^d)$. There exist $A, B \geq 1$, independent of α and p , such that for every $\alpha \in \mathbb{N}^d$ and $1 \leq p \leq \infty$,*

$$A^{-1}B^{-|\alpha|} \exp \mathcal{Q}(\alpha) \leq \|\partial^\alpha f\|_{L^p} \leq AB^{|\alpha|} \exp \mathcal{Q}(\alpha). \quad (1.6)$$

In particular,

$$\log \|\partial^\alpha f\|_{L^p} = \frac{1}{2} \sum_{i,j=1}^d \min(\lambda_i, \lambda_j) \alpha_i \alpha_j + O(|\alpha| + 1). \quad (1.7)$$

If $\alpha^{(n)}/n \rightarrow a \in [0, \infty)^d$, then

$$\frac{1}{n^2} \log \left\| \partial^{\alpha^{(n)}} f \right\|_{L^p} \longrightarrow \frac{1}{2} a^\top M a. \quad (1.8)$$

The limit remains valid if p varies with n in $[1, \infty]$.

This is a two-sided result. In particular, the upper bound cannot be improved by replacing the quadratic form with a strictly smaller value on any nonnegative derivative direction.

1.3 The Fourier statement behind the derivative law

For $s \in [0, \infty)^d$ define

$$\mathcal{H}(s) = \int_0^\infty \max\{0, s_1 - \lambda_1 u, \dots, s_d - \lambda_d u\} du, \quad S = \max_j s_j. \quad (1.9)$$

Let $\log_+ x = \max(0, \log x)$ for $x > 0$, and set $\log_+ 0 = 0$. The characteristic function of (1.2) is

$$\Phi(t) = \mathbb{E} e^{it \cdot Y} = \prod_{k=0}^\infty \operatorname{sinc} \left(\sum_{j=1}^d t_j q_j^k \right), \quad \operatorname{sinc} z = \begin{cases} \sin z / z, & z \neq 0, \\ 1, & z = 0. \end{cases} \quad (1.10)$$

For a logarithmic scale vector s , put

$$B_s = \prod_{j=1}^d [e^{s_j}, 2e^{s_j}]. \quad (1.11)$$

Averages on this box are with respect to its normalized Lebesgue measure.

Theorem 1.2 (Uniform logarithmic-box envelope). *There are constants $C_+, C_- > 0$ such that*

$$\log |\Phi(t)| \leq -\mathcal{H}(\log_+ |t_1|, \dots, \log_+ |t_d|) + C_+(1 + \max_j \log_+ |t_j|), \quad (1.12)$$

$$\frac{1}{\operatorname{Vol} B_s} \int_{B_s} \log |\Phi(t)| dt \geq -\mathcal{H}(s) - C_-(1 + S). \quad (1.13)$$

The value $-\infty$ at a zero is permitted in the first inequality. The integral in the second inequality is finite. Consequently,

$$\log \sup_{t \in B_s} |\Phi(t)| = -\mathcal{H}(s) + O(1 + S), \quad (1.14)$$

uniformly in s . More strongly, for every $0 < \delta < 1$ there is C_δ such that

$$\frac{\operatorname{Vol}\{t \in B_s : \log |\Phi(t)| \geq -\mathcal{H}(s) - C_\delta(1 + S)\}}{\operatorname{Vol} B_s} \geq 1 - \delta. \quad (1.15)$$

The box theorem is stronger than a fixed-ray statement: the relative sizes of the coordinates may vary exponentially with the scale. Its proof controls cancellation directly; it does not assume that frequencies avoid every zero by a prescribed distance.

1.4 The signed resonance in one formula

Theorem 1.3 (A doubling at opposite parameters). *Fix $0 < r < 1$ and $\lambda = -\log r$. The pair (Y_r, Y_{-r}) has a smooth compactly supported density f_0 . For every $a, b \in \mathbb{N}$ and every $1 \leq p \leq \infty$,*

$$\log \left\| \partial_1^a \partial_2^b f_0 \right\|_{L^p} = \lambda(a+b)^2 + O(a+b+1). \quad (1.16)$$

If one incorrectly extended [theorem 1.1](#) to the repeated modulus r, r , one would obtain $\lambda(a+b)^2/2$, exactly half the correct leading term.

2 Support, convergence, and elementary Fourier facts

2.1 The random series and its support

Let $v_k = (q_1^k, \dots, q_d^k)^\top$. Since $\sum_k \|v_k\| < \infty$, the random series [\(1.2\)](#) converges absolutely for every digit sequence. Its support is

$$K = \sum_{k \geq 0} [-v_k, v_k] = \left\{ \sum_{k \geq 0} u_k v_k : -1 \leq u_k \leq 1 \right\}. \quad (2.1)$$

Indeed, the map from the compact product $[-1, 1]^\mathbb{N}$ to $\mathbb{R}^d$ is continuous: a sufficiently long finite prefix controls the image to any prescribed accuracy, uniformly in the remaining digits. The product uniform measure has full support. Its pushforward therefore has support exactly the image in [\(2.1\)](#).

The set K is compact and convex. The first d columns form a Vandermonde matrix,

$$\det[v_0 \cdots v_{d-1}] = \prod_{i < j} (q_j - q_i) \neq 0. \quad (2.2)$$

Hence K has nonempty interior. The first d summands already have a bounded uniform density on a parallelepiped, and convolution with the remaining probability measure gives a bounded density for the full law.

We will only need the elementary support box

$$K \subseteq \prod_{j=1}^d \left[-\frac{1}{1-|q_j|}, \frac{1}{1-|q_j|} \right], \quad W := \prod_{j=1}^d \frac{2}{1-|q_j|} > 1. \quad (2.3)$$

The exact zonoid volume from the repository is not needed for any proof.

2.2 Infinite products and their tails

Independence proves [\(1.10\)](#) first for finite products. For fixed t , $A_k(t) := \sum_j t_j q_j^k$ decays geometrically, so $\sum_k A_k(t)^2 < \infty$. The Taylor expansion $\log \operatorname{sinc} u = -u^2/6 + O(u^4)$ near zero gives convergence of the tail to a nonzero value. Thus a zero of Φ comes from a finite sinc factor. On a compact frequency set only finitely many factors can vanish, so the zero set is locally a finite union of affine hyperplanes.

For real u ,

$$|\operatorname{sinc} u| \leq \min(1, |u|^{-1}), \quad (2.4)$$

with the usual interpretation at zero. Also, for $|u| \leq 1$,

$$0 \leq -\log \operatorname{sinc} u \leq \frac{u^2}{5}. \quad (2.5)$$

To verify the latter, use $\operatorname{sinc} u \geq 1 - u^2/6$ and $-\log(1 - z) \leq z/(1 - z)$ for $0 \leq z \leq 1/6$.

Lemma 2.1 (Uniform interval average of the sine logarithm). *For every $b_0 > 0$, every real interval I of length $b \geq b_0$ satisfies*

$$\frac{1}{b} \int_I \log |\sin u| \, du \geq - \left(1 + \frac{2\pi}{b_0}\right) \log 2. \quad (2.6)$$

Proof. The nonnegative, π -periodic function $h(u) = -\log |\sin u|$ has integral $\pi \log 2$ on one period. An interval of length b is covered by at most $b/\pi + 2$ periods. Integrating h over that cover and dividing by b gives the result. The singularities at integer multiples of π are integrable. $\square$

The identity for the period integral follows, for example, by writing $I = \int_0^{\pi/2} \log \sin u \, du$, observing that the same integral is obtained with cosine, and using $\sin u \cos u = \sin(2u)/2$ to obtain $2I = I - (\pi/2) \log 2$. This keeps the averaging argument independent of any Diophantine assumptions on the parameters.

3 Uniform Fourier upper bounds: only finitely many cancellations matter

Fix $t \in \mathbb{R}^d$. Terms with $t_j = 0$ may be omitted. Put

$$E_j(k) = |t_j| e^{-\lambda_j k}, \quad m(k) = \max(0, \log E_1(k), \dots, \log E_d(k)), \quad S = \max_j \log_+ |t_j|. \quad (3.1)$$

The function $m(u)$ for real $u \geq 0$ is exactly the integrand in $\mathcal{H}(\log_+ |t_1|, \dots, \log_+ |t_d|)$. Replacing a negative logarithm by zero does not change that maximum, because the separate zero term is present.

Lemma 3.1 (Bounded transition set). *Call an integer $k \geq 0$ bad if some pair of nonzero terms satisfies*

$$(2d)^{-1} \leq E_i(k)/E_j(k) \leq 2d.$$

The number of bad integers is at most

$$B_0 = \sum_{i < j} \left(1 + \frac{2 \log(2d)}{|\lambda_i - \lambda_j|}\right). \quad (3.2)$$

At every other integer,

$$|A_k(t)| \geq \frac{1}{2} \max_j E_j(k). \quad (3.3)$$

Proof. For each pair, $\log(E_i(k)/E_j(k))$ is affine in k , with nonzero slope $\lambda_j - \lambda_i$. Its membership in $[-\log(2d), \log(2d)]$ confines k to an interval of length $2 \log(2d)/|\lambda_i - \lambda_j|$. The number of integers in such an interval is at most its length plus one. Summing over pairs proves (3.2). Outside the bad set the largest magnitude is unique and every other magnitude is smaller by a factor exceeding $2d$. Their sum is less than half the largest. The reverse triangle inequality proves (3.3), regardless of signs of either t_j or q_j^k . $\square$

This elementary observation is the uniformity mechanism. A direction may move arbitrarily with frequency, but it can move the transition intervals without increasing their total length. Their contribution is at most linear in the largest logarithmic frequency, whereas the main envelope is quadratic.

Proof of (1.12). For a good k with $m(k) > 0$, (2.4) and (3.3) give

$$\log |\operatorname{sinc} A_k(t)| \leq -m(k) + \log 2.$$

All other factors have logarithm at most zero. The number of integers with $m(k) > 0$ is at most $S/\lambda_{\min} + 1$, where $\lambda_{\min} = \min_j \lambda_j$. Each discarded bad integer loses at most S . Consequently,

$$\log |\Phi(t)| \leq -\sum_{k \geq 0} m(k) + B_0 S + \log 2 \left(\frac{S}{\lambda_{\min}} + 1 \right). \quad (3.4)$$

The function m is decreasing and nonnegative, so

$$\mathcal{H}(\log_+ |t|) \leq \sum_{k \geq 0} m(k) \leq \mathcal{H}(\log_+ |t|) + S. \quad (3.5)$$

Substitution proves the claim. One permissible constant is

$$C_+ = \log 2 + B_0 + \frac{\log 2}{\lambda_{\min}}. \quad (3.6)$$

At a zero of the product the inequality is automatic. $\square$

Corollary 3.2 (Uniform superpolynomial decay). *If $T = \|t\|_\infty \geq 1$ and $\lambda_{\max} = \max_j \lambda_j$, then*

$$|\Phi(t)| \leq \exp \left[-\frac{(\log T)^2}{2\lambda_{\max}} + C_+(1 + \log T) \right]. \quad (3.7)$$

In particular, every polynomial multiple of Φ is integrable. Fourier inversion gives $f \in C_c^\infty(\mathbb{R}^d)$.

Proof. A coordinate with logarithmic size $S = \log T$ contributes the triangle $\int_0^\infty (S - \lambda_j u)_+ du = S^2/(2\lambda_j)$ to the maximum. This is at least $S^2/(2\lambda_{\max})$. The exponential of a negative quadratic in $\log T$ decays faster than every power of T . Apply Fourier inversion and differentiate under the integral. Compact support follows from (2.1). $\square$

4 Matching Fourier lower bounds by logarithmic averaging

Let $R_1, \dots, R_d$ be independent uniforms on $[1, 2]$, and set $t_j = e^{s_j} R_j$. This is uniform on B_s . For each integer k , define

$$b_k = \max_j e^{s_j - \lambda_j k}, \quad m_k = \log b_k. \quad (4.1)$$

Choose an index $j(k)$ attaining the maximum. Conditional on all other R_i , the random argument $A_k(t)$ is uniform on an interval of length b_k . Its orientation may reverse for signed parameters, which has no effect on the normalized integral. Moreover,

$$|A_k(t)| \leq 2d b_k. \quad (4.2)$$

Lemma 4.1 (One-factor mean bound). *If $b_k \geq (2d)^{-1}$, then*

$$\mathbb{E} \log |\operatorname{sinc} A_k(t)| \geq -(m_k)_+ - C_d, \quad C_d = (1 + 4\pi d) \log 2 + \log(2d). \quad (4.3)$$

Proof. Apply [theorem 2.1](#) conditionally, with $b_0 = (2d)^{-1}$. Also $\log |A_k(t)| \leq \log(2d) + m_k$ wherever $A_k(t) \neq 0$. The single point $A_k(t) = 0$ has conditional measure zero and is removable for the sinc function. Subtracting the denominator logarithm gives $\mathbb{E} \log |\text{sinc } A_k| \geq -C_d - m_k$. Replacing m_k by $(m_k)_+$ only weakens this inequality. $\square$

Proof of (1.13). The integers with $b_k \geq (2d)^{-1}$ form an initial segment of length at most

$$1 + \frac{S + \log(2d)}{\lambda_{\min}}.$$

The total of $(m_k)_+$ is at most $\mathcal{H}(s) + S$ by (3.5). After the initial segment, $b_{k+\ell} \leq b_k e^{-\lambda_{\min} \ell}$ and $2db_k < 1$. Equations (2.5) and (4.2) bound the negative logarithm of the remaining product by

$$\frac{1}{5(1 - e^{-2\lambda_{\min}})}, \quad (4.4)$$

uniformly in the box. Summing (4.3) and the tail estimate yields

$$\mathbb{E} \log |\Phi(t)| \geq -\mathcal{H}(s) - S - C_d \left(1 + \frac{S + \log(2d)}{\lambda_{\min}} \right) - \frac{1}{5(1 - e^{-2\lambda_{\min}})}. \quad (4.5)$$

This has the required form. Every $-\log |\text{sinc } A_k|$ is nonnegative, so Tonelli's theorem applies to the negative logarithms. The displayed finite bound proves integrability and justifies the infinite-product average. $\square$

Completion of theorem 1.2. For $t \in B_s$, its actual logarithmic scale lies coordinatewise between s and $s + \log 2$. Monotonicity of $\mathcal{H}$ and (1.12) give $\log |\Phi(t)| \leq -\mathcal{H}(s) + C_0(1 + S)$ for a fixed C_0 . The average lower bound implies the existence of a point with $\log |\Phi(t)| \geq -\mathcal{H}(s) - C_-(1 + S)$, proving (1.14). For the measure statement, set

$$D(t) = -\log |\Phi(t)| - \mathcal{H}(s) + C_0(1 + S) \geq 0.$$

Its expectation is at most $(C_0 + C_-)(1 + S)$. Markov's inequality gives $D(t) \leq (C_0 + C_-)(1 + S)/\delta$ outside a set of relative measure at most δ . Rearranging proves (1.15). $\square$

Remark 4.2 (What averaging does and does not remove). The lower bound does not hold at all frequencies: exact zero hyperplanes remain, and arbitrarily small neighborhoods of them contain arbitrarily small values. Averaging replaces a vague exclusion of zeros with a precise and uniform statement about the proportion of each box on which the sharp scale is attained. No independence between different sinc factors is assumed; the same d box variables occur in every factor.

5 The convex dual is a Brownian covariance matrix

Relabel coordinates temporarily so that

$$0 < \lambda_1 < \lambda_2 < \cdots < \lambda_d. \quad (5.1)$$

The statement in the original coordinates is recovered by permutation. For $a \in [0, \infty)^d$, let

$$A_j = \sum_{i=j}^d a_i, \quad A_{d+1} = 0, \quad s^* = Ma. \quad (5.2)$$

Theorem 5.1 (Exact positive-orthant conjugacy). *For every $a \in [0, \infty)^d$,*

$$\sup_{s \in [0, \infty)^d} \{a \cdot s - \mathcal{H}(s)\} = \frac{1}{2} a^\top M a. \quad (5.3)$$

The supremum is attained at $s^ = M a$, and*

$$\mathcal{H}(M a) = \frac{1}{2} a^\top M a. \quad (5.4)$$

Proof. For arbitrary s , integrate the j th line over the interval $[A_{j+1}, A_j]$. Since the maximum in (1.9) is at least that line,

$$\begin{aligned} \mathcal{H}(s) &\geq \sum_{j=1}^d \int_{A_{j+1}}^{A_j} (s_j - \lambda_j u) du \\ &= a \cdot s - \frac{1}{2} \sum_{j=1}^d \lambda_j (A_j^2 - A_{j+1}^2). \end{aligned} \quad (5.5)$$

The subtracted term is

$$\frac{1}{2} \sum_{j=1}^d (\lambda_j - \lambda_{j-1}) A_j^2 = \frac{1}{2} \sum_{i,j} \min(\lambda_i, \lambda_j) a_i a_j, \quad \lambda_0 = 0. \quad (5.6)$$

This proves the upper bound on the supremum.

For equality, note that

$$s_j^* = \sum_{i < j} \lambda_i a_i + \lambda_j \sum_{i \geq j} a_i, \quad s_1^* = \lambda_1 A_1, \quad s_j^* - s_{j-1}^* = (\lambda_j - \lambda_{j-1}) A_j. \quad (5.7)$$

The lines with indices j and $j-1$ cross at $u = A_j$. If $A_{j+1} \leq u \leq A_j$, then the j th line dominates every smaller-index line: each adjacent difference has the sign of $A_h - u \geq 0$ for $h \leq j$. It dominates every larger-index line because those adjacent differences have the sign of $A_h - u \leq 0$ for $h > j$. The first line is nonnegative up to A_1 ; on the selected intervals the maximum is therefore nonnegative. After A_1 all lines are nonpositive. Thus the maximum equals exactly the selected line on each prescribed interval. Equality holds in (5.5). Finally $a \cdot M a = 2Q(a)$, so (5.4) follows. Zero entries of a merely produce intervals of length zero and require no separate argument. $\square$

5.1 Why the word “Brownian” is exact

Let L be the lower-triangular matrix whose entries on and below the diagonal are one. Then

$$M = L \operatorname{diag}(\lambda_1, \lambda_2 - \lambda_1, \dots, \lambda_d - \lambda_{d-1}) L^\top. \quad (5.8)$$

In particular,

$$\det M = \lambda_1 \prod_{j=2}^d (\lambda_j - \lambda_{j-1}) > 0. \quad (5.9)$$

Independent centered increments with those variances have cumulative-sum covariance M . Equivalently, a standard Brownian motion sampled at times λ_j has this covariance, by the independent-increment variance calculation. This is an identification of a finite matrix, not an assertion that the original Fabius law is Gaussian.

The useful expression for the smoothness cost is

$$\mathcal{Q}(\alpha) = \frac{1}{2} \sum_{j=1}^d (\lambda_j - \lambda_{j-1}) \left(\sum_{i=j}^d \alpha_i \right)^2. \quad (5.10)$$

Each scale increment charges the square of the total derivative order still present at that scale. This explains the cross terms.

Remark 5.2 (The envelope itself is not globally one quadratic). Equation (5.3) is a conjugacy on the nonnegative orthant. It does not say that $\mathcal{H}(s) = s^\top M^{-1} s / 2$ for every nonnegative s . Some lines may be inactive, and $\mathcal{H}$ is piecewise quadratic. The positive orthant of dual variables selects the chamber structure encoded by (5.7). Confusing the envelope with a single quadratic would give wrong Fourier predictions in inactive-coordinate regimes.

6 Proof of the sharp derivative theorem

6.1 An integrable Fourier moment bound

Define

$$I_\alpha = \int_{\mathbb{R}^d} \prod_j |t_j|^{\alpha_j} |\Phi(t)| dt. \quad (6.1)$$

Lemma 6.1 (Uniform moment estimate). *There are $A_1, B_1 \geq 1$ such that*

$$I_\alpha \leq A_1 B_1^{|\alpha|} e^{\mathcal{Q}(\alpha)} \quad (6.2)$$

for every multi-index. The same bound holds for real nonnegative exponents.

Proof. Use (1.12), split into orthants, and then split each coordinate at $|t_j| = 1$. On a piece with active set J , set $|t_j| = e^{s_j}$ for $j \in J$, and set $s_j = 0$ otherwise. Integrating the inactive coordinates gives factors $1/(\alpha_j + 1) \leq 1$. The Jacobian on an active coordinate contributes e^{s_j} . Since $S \leq \sum_j s_j$, the remaining integrand is at most

$$e^{C_+} \exp((\alpha + (1 + C_+) \mathbf{1}) \cdot s - \mathcal{H}(s)).$$

Put $c = 2 + C_+$ and $\gamma = \alpha + c \mathbf{1}$. By [theorem 5.1](#),

$$(\alpha + (1 + C_+) \mathbf{1}) \cdot s - \mathcal{H}(s) \leq \mathcal{Q}(\gamma) - \sum_j s_j.$$

The integral of the final exponential over any active face is one. There are 2^d orthants and 2^d active-set choices, hence

$$I_\alpha \leq 4^d e^{C_+} \exp \mathcal{Q}(\alpha + c \mathbf{1}). \quad (6.3)$$

Expanding the quadratic,

$$\mathcal{Q}(\alpha + c \mathbf{1}) = \mathcal{Q}(\alpha) + c \alpha^\top M \mathbf{1} + c^2 \mathcal{Q}(\mathbf{1}).$$

The last two terms are bounded by a fixed constant plus a fixed constant times $|\alpha|$. This proves (6.2). Notice that no loss of order $|\alpha| \log |\alpha|$ was introduced. $\square$

6.2 The upper bound in physical space

Our Fourier convention is $\widehat{f}(t) = \int e^{it \cdot x} f(x) dx$. Fourier inversion gives

$$\partial^\alpha f(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} (-it)^\alpha e^{-it \cdot x} \Phi(t) dt. \quad (6.4)$$

Therefore $\|\partial^\alpha f\|_\infty \leq (2\pi)^{-d} I_\alpha$. Every derivative is supported in K , so

$$\|\partial^\alpha f\|_{L^p} \leq W^{1/p} \|\partial^\alpha f\|_\infty \leq W \|\partial^\alpha f\|_\infty. \quad (6.5)$$

This proves the upper half of [theorem 1.1](#), uniformly in p .

6.3 A frequency witness gives the lower bound

For any compactly supported smooth g with support in the box (2.3), Hölder's inequality gives

$$|\widehat{g}(t)| \leq \|g\|_1 \leq W^{1-1/p} \|g\|_p \leq W \|g\|_p. \quad (6.6)$$

Apply this to $g = \partial^\alpha f$. For every t ,

$$W \|\partial^\alpha f\|_p \geq \prod_j |t_j|^{\alpha_j} |\Phi(t)|. \quad (6.7)$$

Choose $s = M\alpha$. The average lower bound in [theorem 1.2](#) supplies a point $t \in B_s$ with

$$\log |\Phi(t)| \geq -\mathcal{H}(s) - C_-(1 + \max_j s_j).$$

Since $t_j \geq e^{s_j}$, [theorem 5.1](#) yields

$$\begin{aligned} \log \|\partial^\alpha f\|_p &\geq \alpha \cdot s - \mathcal{H}(s) - C_-(1 + \max_j s_j) - \log W \\ &\geq \mathcal{Q}(\alpha) - C_-(1 + \lambda_{\max}|\alpha|) - \log W. \end{aligned} \quad (6.8)$$

This is the lower half of [theorem 1.1](#). The scaling limit (1.8) follows by dividing by n^2 and using continuity of the quadratic form. Uniformity in p has been retained throughout.

6.4 The normalized family and independent marginals

If $X = b + DY$ with $D = \text{diag}(c_1, \dots, c_d)$ and all $c_j > 0$, then

$$\|\partial^\alpha f_X\|_p = (\det D)^{1/p-1} \prod_j c_j^{-\alpha_j} \|\partial^\alpha f\|_p. \quad (6.9)$$

Taking logarithms proves the normalization claim following (1.4).

For comparison, replace the shared digits by independent digit sequences in the d coordinates. The joint density is a product, so its L^p mixed norm is the product of its one-dimensional derivative norms. Its leading cost is only

$$\mathcal{Q}_{\text{ind}}(\alpha) = \frac{1}{2} \sum_j \lambda_j \alpha_j^2. \quad (6.10)$$

Thus common digits add exactly the cross cost

$$\mathcal{Q}(\alpha) - \mathcal{Q}_{\text{ind}}(\alpha) = \sum_{i < j} \min(\lambda_i, \lambda_j) \alpha_i \alpha_j. \quad (6.11)$$

The distinction survives at order $|\alpha|^2$ and cannot be absorbed into normalization constants.

7 Fixed rays, moving directions, and directional derivatives

7.1 A precise resolution of the leading ray envelope

Theorem 7.1 (Ray-window envelope and good frequencies). *Fix $\theta \neq 0$. Let*

$$\lambda_*(\theta) = \min\{\lambda_j : \theta_j \neq 0\}. \quad (7.1)$$

For $R \geq 2$,

$$\log \sup_{R \leq r \leq 2R} |\Phi(r\theta)| = -\frac{(\log R)^2}{2\lambda_*(\theta)} + O_\theta(\log R). \quad (7.2)$$

There is also the pointwise upper bound with the same leading term and an $O_\theta(\log R)$ error throughout the window. For each $0 < \delta < 1$, a set of $r \in [R, 2R]$ of measure at least $(1 - \delta)R$ satisfies the matching lower bound with error $O_{\theta, \delta}(\log R)$.

Proof. Set $a_k = \sum_j \theta_j q_j^k$. The term with largest active modulus is unique, so for all sufficiently large k ,

$$c_1 e^{-\lambda_* k} \leq |a_k| \leq c_2 e^{-\lambda_* k} \quad (7.3)$$

for constants $c_1, c_2 > 0$. In particular, only finitely many a_k vanish; those factors are identically one. If $L = \log R$, the remaining factors obey

$$\sum_{k: a_k \neq 0} \log_+(R|a_k|) = \frac{L^2}{2\lambda_*} + O_\theta(L). \quad (7.4)$$

For the finite initial segment the contribution is $O_\theta(L)$. For the tail, compare its logarithms using (7.3) to the arithmetic progression $L - \lambda_* k$ and sum its positive part.

The pointwise bound (2.4) proves the desired upper estimate. For the average lower estimate, let r be uniform on $[R, 2R]$. Whenever $R|a_k| \geq 1/2$, the argument ra_k is uniform on an interval of length at least $1/2$. Theorem 2.1, and $|ra_k| \leq 2R|a_k|$, give a lower mean bound $-\log_+(R|a_k|) - C$ for that factor. There are $O_\theta(L)$ such factors. For all remaining sufficiently late factors, (2.5) and (7.3) give a total $O_\theta(1)$ loss. Increasing the constant handles the finitely many initial indices and bounded R . Thus the average logarithm is bounded below by the right side of (7.2). The supremum and relative-measure conclusions follow by the same nonnegative-deficiency argument as in section 4. $\square$

For positive parameters, the active q_* in the repository conjecture is $e^{-\lambda_*}$. Hence (7.2) proves its proposed leading coefficient in a precisely defined envelope sense. No rational independence of the logarithms is needed for this conclusion. The arithmetic dependence of finer oscillations remains a separate question.

Corollary 7.2 (Sharp isotropic shell envelope). *Under (1.3),*

$$\log \sup_{R \leq \|t\|_\infty \leq 2R} |\Phi(t)| = -\frac{(\log R)^2}{2\lambda_{\max}} + O(\log R). \quad (7.5)$$

Proof. The upper bound is (3.7). For the lower bound, take the coordinate axis corresponding to $\lambda_{\max}$ in theorem 7.1. $\square$

7.2 Physical directional derivatives select the opposite extreme

For $v \neq 0$, write $D_v = \sum_j v_j \partial_j$ and define

$$\lambda^\sharp(v) = \max\{\lambda_j : v_j \neq 0\}. \quad (7.6)$$

Theorem 7.3 (Directional derivative spectrum). *For each fixed $v \neq 0$ and every $1 \leq p \leq \infty$,*

$$\log \|D_v^n f\|_p = \frac{\lambda^\sharp(v)}{2} n^2 + O_v(n+1), \quad (7.7)$$

uniformly in p .

Proof. Expand D_v^n by the multinomial theorem. Every occurring multi-index is supported on $\{j : v_j \neq 0\}$, so $Q(\alpha) \leq \lambda^\sharp(v) n^2/2$. The upper bound follows from [theorem 1.1](#); the sum of the absolute multinomial coefficients is $(\sum_j |v_j|)^n$, contributing only $O_v(n)$ to its logarithm.

For the lower bound, choose the unique j in that support with largest λ_j , and set $s = nMe_j$. Then $s_j = \lambda_j n$, and for every active $i \neq j$ one has $s_i = \lambda_i n < \lambda_j n$. Uniformly for $t \in B_s$,

$$|v \cdot t| \geq \frac{|v_j|}{2} e^{\lambda_j n}$$

for all sufficiently large n . Coordinates with greater λ_i have $v_i = 0$ and cause no obstruction. Choose a point in that box satisfying the Fourier lower bound. Since $\mathcal{H}(s) = \lambda_j n^2/2$, the Fourier transform identity for $D_v^n f$ and [\(6.6\)](#) give the matching lower bound. The finitely many smaller orders are absorbed into the constant; $D_v^n f$ is not identically zero, since its Fourier transform $(-iv \cdot t)^n \Phi(t)$ is nonzero at some sufficiently small t . $\square$

A Fourier ray is governed by the *smallest* active logarithmic rate; a physical directional derivative is governed by the *largest* one. These are dual phenomena, not competing answers to the same question. The derivative spectrum is constant on the successive coordinate strata obtained by deleting the fastest remaining direction.

8 Opposite parameters: exact parity factorization and noncommuting limits

8.1 An exact factorization at a repeated modulus

Let $q_1 = r$, $q_2 = -r$, and $\lambda = -\log r$. Split the digits into their independent even and odd subsequences:

$$E = \sum_{k \geq 0} r^{2k} V_{2k}, \quad O = \sum_{k \geq 0} r^{2k} V_{2k+1}. \quad (8.1)$$

Both E and O have the one-dimensional density h with parameter r^2 , and

$$Y_r = E + rO, \quad Y_{-r} = E - rO. \quad (8.2)$$

Therefore

$$f_0(x, y) = \frac{1}{2r} h\left(\frac{x+y}{2}\right) h\left(\frac{x-y}{2r}\right). \quad (8.3)$$

In Fourier space this is

$$\Phi_{r,-r}(t_1, t_2) = \phi_{r^2}(t_1 + t_2) \phi_{r^2}(r(t_1 - t_2)), \quad (8.4)$$

where ϕ_{r^2} denotes the one-dimensional centered product.

Proof of theorem 1.3. The one-dimensional case of theorem 1.1, with logarithmic rate 2λ , gives

$$\log \|h^{(k)}\|_\infty = \lambda k^2 + O(k + 1).$$

In the coordinates (E, O) , $\partial_x = \frac{1}{2}\partial_E + \frac{1}{2\tau}\partial_O$ and $\partial_y = \frac{1}{2}\partial_E - \frac{1}{2\tau}\partial_O$. A derivative of total order $N = a + b$ expands into terms involving $h^{(k)}(E)h^{(N-k)}(O)$. Their quadratic costs are

$$\lambda(k^2 + (N - k)^2) \leq \lambda N^2.$$

The total coefficient sum is at most a fixed constant to the power N . Together with compact support this proves the upper estimate for all p .

For the lower estimate, set $t_1 = t_2 = \tau$. Equation (8.4) reduces to $\phi_{\tau,2}(2\tau)$, while the mixed monomial multiplier is $|\tau|^N$. By the one-dimensional ray-window bound, there is $\tau \in [R, 2R]$ with

$$\log |\phi_{\tau,2}(2\tau)| \geq -\frac{(\log R)^2}{4\lambda} - C \log R.$$

Choose $\log R = 2\lambda N$ and apply the compact-support Fourier lower bound. The resulting exponent is $2\lambda N^2 - \lambda N^2 + O(N) = \lambda N^2 + O(N)$. This also handles mixed derivatives whose physical-space expansion contains cancellations: the Fourier witness avoids estimating that expansion from below. $\square$

8.2 Continuity at each fixed derivative order

Lemma 8.1 (Fixed-order parameter continuity). *Let $q^{(m)} \rightarrow q^{(0)}$ in $((-1, 1) \setminus \{0\})^d$, with the coordinates of $q^{(0)}$ distinct as signed numbers. Repeated moduli at the limit are allowed. For all sufficiently large m the corresponding densities exist, and for every fixed α they satisfy*

$$\partial^\alpha f_{q^{(m)}} \longrightarrow \partial^\alpha f_{q^{(0)}} \quad \text{in every } L^p, \quad 1 \leq p \leq \infty. \quad (8.5)$$

Proof. Work in a compact neighborhood of $q^{(0)}$ on which no parameter is zero, no signed coordinates coincide, and every modulus is bounded above by a number smaller than one. For each fixed block of d consecutive digit vectors, its matrix is a nonzero diagonal matrix times a Vandermonde matrix. Its least singular value has a positive minimum on this neighborhood. Consequently at least one argument in that block has magnitude at least a fixed positive multiple of $\|t\|$. Its sinc factor is $O((1 + \|t\|)^{-1})$ uniformly in the parameters. Using m disjoint blocks gives a uniform $O_m((1 + \|t\|)^{-m})$ bound. For each fixed t , the products converge continuously in the parameters by uniform geometric control of their tails. Choose $m > d + |\alpha|$ and apply dominated convergence to the Fourier inversion formula for the difference. This gives uniform convergence of the derivatives. Their supports lie in a common compact box, so the convergence also holds in every finite L^p . $\square$

Corollary 8.2 (Noncommuting small-separation and large-order limits). *Let f_ε be the density at $(r, -re^{-\varepsilon})$, for $\varepsilon \geq 0$, and put $G_n(\varepsilon) = n^{-2} \log \|\partial_1^n f_\varepsilon\|_p$. For every fixed $p \in [1, \infty]$,*

$$\boxed{\lim_{\varepsilon \downarrow 0} \lim_{n \rightarrow \infty} G_n(\varepsilon) = \frac{\lambda}{2}, \quad \lim_{n \rightarrow \infty} \lim_{\varepsilon \downarrow 0} G_n(\varepsilon) = \lambda.} \quad (8.6)$$

Proof. For each $\varepsilon > 0$, theorem 1.1 with $\alpha = (n, 0)$ gives the first inner limit $\lambda/2$. For each fixed n , theorem 8.1 identifies the other inner limit with $G_n(0)$. The resonance law then gives λ . The norms are positive, so taking logarithms is legitimate. $\square$

The singularity is therefore in high-order asymptotics, not in the density or any fixed finite number of its derivatives. This is why compact-parameter continuity cannot justify extending the distinct-modulus formula to a resonant endpoint.

9 Consequences for regularity, convolution, and computation

9.1 The exact global ultradifferentiable scale

A global Gevrey estimate of finite order σ would have the form

$$\|\partial^\alpha f\|_\infty \leq AB^{|\alpha|}(|\alpha|!)^\sigma.$$

For $\alpha = ne_j$, the logarithm of this bound is only $O(n \log n)$, whereas [theorem 1.1](#) gives $\lambda_j n^2/2 + O(n)$. Thus no finite Gevrey order contains f globally. The exact replacement, up to exponential factors in the derivative order, is the anisotropic weight $e^{\mathcal{Q}(\alpha)}$.

The global conclusion does not, by itself, assert failure of a Gevrey bound on *every* interior open subset. Norms on the whole support do not locate the points where large derivatives occur. Local regularity is a separate problem listed in [section 12](#).

9.2 Convolution powers divide the quadratic cost

Theorem 9.1 (Reciprocal convolution-power law). *For a fixed integer $m \geq 1$, let f^{*m} be the density of the sum of m independent copies of Y . Then*

$$\log \|\partial^\alpha (f^{*m})\|_p = \frac{1}{m} \mathcal{Q}(\alpha) + O_m(|\alpha| + 1), \quad 1 \leq p \leq \infty, \quad (9.1)$$

uniformly in α and p .

Proof. The characteristic function is Φ^m , so both the upper envelope and the averaged lower envelope are multiplied by m . The relevant conjugacy is

$$\sup_{s \geq 0} \{\alpha \cdot s - m\mathcal{H}(s)\} = \frac{1}{m} \mathcal{Q}(\alpha), \quad s^* = \frac{1}{m} M\alpha.$$

This follows either by homogeneity or by [theorem 5.1](#) applied to α/m . The Fourier-moment and compact-support witness arguments from [section 6](#) now apply verbatim with those modified quantities. All remaining losses are linear in $|\alpha|$ for fixed m . $\square$

Finite convolution improves the exact quadratic coefficient but does not bring the density into any finite Gevrey class. Averaging rather than summing the m copies makes only a fixed diagonal rescaling and leaves the displayed quadratic coefficient unchanged.

9.3 Quantitative boundary flatness

Every derivative of f vanishes on ∂K , because it is continuous on $\mathbb{R}^d$ and zero outside K . The directional estimate makes this qualitative flatness quantitative.

Corollary 9.2 (A logarithmic flatness estimate). *Fix $v \neq 0$ and let $\Lambda = \lambda^\sharp(v)$. Uniformly over $x_0 \in \partial K$, as $h \downarrow 0$ and $L = \log(1/h)$,*

$$\log |f(x_0 + hv)| \leq -\frac{L^2}{2\Lambda} - \frac{L \log L}{\Lambda} + O_v(L). \quad (9.2)$$

At a zero the left side is interpreted as $-\infty$.

Proof. All directional Taylor coefficients at x_0 vanish. For every $n \geq 1$, Taylor's integral remainder and [theorem 7.3](#) give

$$|f(x_0 + hv)| \leq \frac{h^n}{n!} \|D_v^n f\|_\infty \leq A_v \frac{(B_v h)^n}{n!} e^{\Lambda n^2/2}.$$

Take $n = \lfloor (L + \log L)/\Lambda \rfloor$ for sufficiently large L and use $\log n! \geq n \log n - n$. Expanding the resulting expression gives [\(9.2\)](#). The constants are independent of x_0 because the starting derivative bounds are global. $\square$

This is an upper bound for density values, not a full small-deviation asymptotic and not a matching lower bound along every boundary approach. It does not replace the repository's endpoint-mass or jet small-ball investigations.

9.4 A coupled Fourier truncation budget

Let $R_j \geq 1$, $L_j = \log R_j$, and define the rectangular inverse transform

$$f_R(x) = (2\pi)^{-d} \int_{\prod_j [-R_j, R_j]} e^{-it \cdot x} \Phi(t) dt. \quad (9.3)$$

Proposition 9.3 (Derivative-aware rectangular truncation). *There are constants $A, B \geq 1$ and $C > 0$ such that*

$$\|\partial^\alpha(f - f_R)\|_\infty \leq AB^{|\alpha|} e^{\mathcal{Q}(\alpha)} \sum_{j=1}^d \exp \left[-\frac{(L_j - (M\alpha)_j - C)_+^2}{2\lambda_j} \right]. \quad (9.4)$$

Proof. Use the union bound for the complement of the rectangle. On $|t_j| > R_j$, for each integer $m \geq 0$,

$$\int |t^\alpha \Phi(t)| \mathbf{1}_{\{|t_j| > R_j\}} dt \leq R_j^{-m} I_{\alpha + me_j}.$$

By [theorem 6.1](#), its logarithmic bound, apart from a constant and $|\alpha| \log B + \mathcal{Q}(\alpha)$, is

$$\frac{\lambda_j m^2}{2} - m(L_j - (M\alpha)_j - \log B).$$

Choose the nearest nonnegative integer to the real minimizer. The rounding loss is at most $\lambda_j/8$. Absorb this bounded loss and the Fourier normalization into A , sum over j , and put $C = \log B$. $\square$

The preferred logarithmic frequency scale for a derivative is $M\alpha$, not the coordinatewise scale $(\lambda_j \alpha_j)_j$ that independent marginals would suggest. The proposition is a rigorous analytic truncation estimate. The accompanying code evaluates Fourier products but does not implement a fully interval-certified multidimensional quadrature scheme.

10 Worked examples and normalization checks

10.1 Two parameters: an explicit piecewise-quadratic envelope

Suppose $0 < \lambda_1 < \lambda_2$. Directly comparing the two lines gives

$$\mathcal{H}(s_1, s_2) = \begin{cases} \frac{s_1^2}{2\lambda_1}, & s_2 \leq s_1, \\ \frac{s_1^2}{2\lambda_1} + \frac{(s_2 - s_1)^2}{2(\lambda_2 - \lambda_1)}, & s_1 < s_2 < \frac{\lambda_2}{\lambda_1} s_1, \\ \frac{s_2^2}{2\lambda_2}, & s_2 \geq \frac{\lambda_2}{\lambda_1} s_1. \end{cases} \quad (10.1)$$

The formulas agree on the interfaces. In the middle region the faster line dominates first and the slower line dominates later. Outside it, one coordinate contributes no active interval to the envelope.

Take $q_1 = 1/2$ and $q_2 = 1/4$. Then

$$M = (\log 2) \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathcal{Q}(a, b) = \frac{\log 2}{2}(a^2 + 2ab + 2b^2). \quad (10.2)$$

In particular,

$$\begin{aligned} \log \|\partial_1^n f\|_p &= \frac{1}{2}(\log 2)n^2 + O(n), \\ \log \|\partial_2^n f\|_p &= (\log 2)n^2 + O(n), \\ \log \|\partial_1^n \partial_2^n f\|_p &= \frac{5}{2}(\log 2)n^2 + O(n). \end{aligned}$$

The independent-marginal analogue of the last line has coefficient $3 \log 2/2$, rather than $5 \log 2/2$.

10.2 Three parameters: the saddle has exact integer geometry

Take $\lambda = (1, 2, 4)$ and $a = (2, 1, 2)$. Then

$$M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 4 \end{pmatrix}, \quad s^* = Ma = (5, 8, 12), \quad \mathcal{Q}(a) = 21. \quad (10.3)$$

The active intervals have lengths $a_3 = 2$, $a_2 = 1$, and $a_1 = 2$, in that order. Their areas are 16, 3, and 2, summing to 21. The dual linear reward is $a \cdot s^* = 42$.

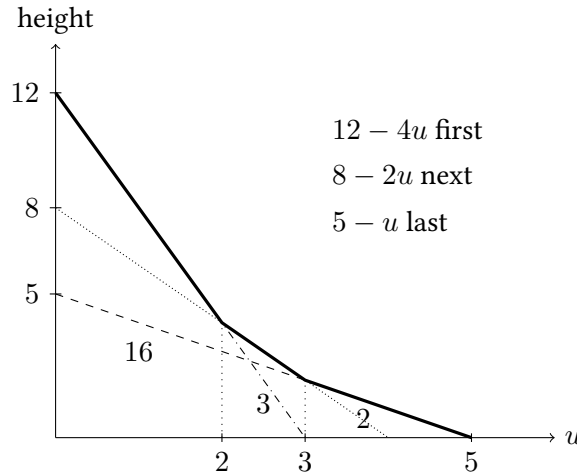


Figure 1: The envelope at the exact saddle $s^* = (5, 8, 12)$. The numbers inside the three regions are their areas, not measured data.

For derivative index $\alpha = k(2, 1, 2)$, the saddle is $k(5, 8, 12)$ and the quadratic cost is $21k^2$. This is the family used in the numerical diagnostics below.

10.3 The classical one-dimensional dyadic check

Arias de Reyna's derivative identity for the normalized up-function up on $[-1, 1]$ gives

$$\left\| \text{up}^{(n)} \right\|_{\infty} = 2^{n(n+1)/2} \quad (10.4)$$

from its translated, signed copies with maximum magnitude one [3, Theorem 4]. In our centered normalization with $q = 1/2$, $Y/2$ has the up-density, hence

$$f(y) = \frac{1}{2} \text{up}(y/2), \quad \left\| f^{(n)} \right\|_{\infty} = 2^{(n^2-n)/2-1}. \quad (10.5)$$

Its logarithmic leading term is $(\log 2)n^2/2$, as required. Equation (10.5) is a consistency check against a classical special case, not a claimed new result.

11 Verification, reproducibility, and the formalization boundary

11.1 Exact finite checks

The script `code/verify.py` uses rational arithmetic for the finite convex geometry. With a fixed seed, it tests 1,200 parameter/order cases in dimensions one through six. Each case checks seven statements: equality $\mathcal{H}(Ma) = \mathcal{Q}(a)$; the identity $a \cdot Ma = 2\mathcal{Q}(a)$; the Fenchel inequality at an independently chosen scale; degree-two homogeneity; the tail-sum formula (5.10); the determinant (5.9); and the isotropic domination $\mathcal{Q}(a) \leq \lambda_{\max}|a|^2/2$.

The envelope integral is computed independently by enumerating pairwise line crossings and integrating the winning line on each interval. Thus the check does not merely evaluate the same quadratic formula twice. It also checks 100 signed-parity coefficient cases, with two exact assertions in each case, and the worked area value 21. All 8,601 exact assertions passed. These finite checks are useful error detectors; the general proofs remain the analytic and algebraic arguments in the article.

11.2 High-precision product diagnostics

The numerical run used Python 3.13.5, mpmath 1.3.0, and 160 decimal digits. Each of five logarithmic boxes and eight ray windows was sampled at 96 reproducibly generated points. These samples are diagnostics, not a proof of the mean inequalities or a numerical optimization certificate.

The omitted product tail is bounded using (2.5): if the next coefficient vector is b and $r = \max_j |q_j|$, then once $\sum_j |b_j| \leq 1$,

$$0 \leq -\log \left| \prod_{k \text{ in tail}} \text{sinc } A_k \right| \leq \frac{(\sum_j |b_j|)^2}{5(1-r^2)}. \quad (11.1)$$

The largest evaluated tail bound in the box and ray runs was below 10^{-90} . This bounds truncation analytically but does *not* enclose floating-point rounding error; the computations are not interval-certified. The signed factorization (8.4) was also tested numerically, with maximum discrepancy in logarithms below 4.6×10^{-101} .

k	$ \alpha $	$\mathcal{Q}(\alpha)$	Sample mean $\log \Phi $	Best log witness
1	5	21	-35.450	9.200
2	10	84	-110.071	64.985
4	20	336	-384.602	300.770
8	40	1344	-1438.377	1274.007
16	80	5376	-5562.780	5233.148

Table 1: The family $\lambda = (1, 2, 4)$, $\alpha = k(2, 1, 2)$. The witness is $\sum_j \alpha_j \log t_j + \log |\Phi(t)| - \log W$, with the omitted-tail loss subtracted. Analytically it lower-bounds the logarithm of every L^p derivative norm at that frequency; the displayed numerical values themselves have not been interval-enclosed.

The files `box_diagnostics.csv` and `ray_diagnostics.csv` preserve more digits and additional normalized errors. The verification report records versions, precision, counts, and the numerical status.

11.3 A feasible proof-assistant decomposition

No Lean or Rocq file is supplied, and none of the new analytic theorems is claimed to be kernel-checked. A useful formalization order would be the following. First prove the finite matrix factorization and the tail-sum quadratic identity. Next formalize the line-envelope integral, the exact conjugacy inequality, and the attaining scale Ma . Those statements have short finite-dimensional dependencies and are independently valuable.

The analytic stage should separate the one-factor logarithmic integral from the infinite product. It needs the integrability of $\log |\sin|$ over compact intervals, the conditional uniform-interval calculation, and Tonelli for nonnegative negative logarithms. The uniform upper estimate then requires only a finite bad-index bound and sum–integral comparison. Fourier inversion, compact support, and Hölder yield the derivative theorem.

Finally, the parity factorization and fixed-order parameter-continuity lemma should be proved independently before connecting them to the noncommuting limits. This ordering prevents a proof from silently reusing the nonresonant estimate at a resonant parameter, where that estimate is false.

12 Further research questions

The main quadratic smoothness problem is settled above under (1.3). The next questions concern finer information or regimes deliberately excluded from the theorem.

Research question 1 (The linear-order derivative profile). For a fixed rational direction $a \geq 0$, determine the behavior of

$$\log \|\partial^{na} f\|_p - n^2 \mathcal{Q}(a)$$

on admissible integer subsequences. Is there a linear term with a limit coefficient, an order- n oscillation, or a dependence on p at that scale? The classical dyadic one-dimensional formula shows that an exact linear term can exist, but it does not determine the coupled case.

Research question 2 (A distributional refinement of the Fourier envelope). Let a frequency be uniform in a box B_{ns} . Study the centered logarithm $\log |\Phi(t)| + n^2 \mathcal{H}(s)$ after division by n or by a smaller fluctuation scale. This asks for genuine joint sine-product statistics, beyond the first-moment argument used here. It is also the proper place to assess the repository’s periodic/quasiperiodic remainder proposal rather than treating it as a consequence of the leading envelope.

Research question 3 (The near-opposite-parameter crossover). For $(r, -re^{-\varepsilon})$, find a uniform formula when the derivative order and ε^{-1} both grow. The two limiting quadratic coefficients are now known exactly. When ε is exponentially small in the order, odd-digit Fourier factors can remain effectively suppressed over a substantial range. Determine the correct transition variables and the resulting interpolating quadratic profile; no crossover formula is proved here.

Research question 4 (Confluent jets). When positive parameters coalesce, divided differences lead to coefficient vectors involving $k^m q^k$. Determine the sharp mixed-derivative law in a nonsingular jet coordinate system, including possible order- $n \log n$ corrections caused by polynomial factors. Taking the limit of M alone is not sufficient, because the coordinate transformation becomes singular.

Research question 5 (Complex parameters and residue-class resonances). For parameters $re^{i\theta_j}$, view the law in its real span and classify how rational phase differences split digits into residue classes.

The pair $r, -r$ is the period-two example. For irrational phases, cancellation is no longer confined by distinct real exponential slopes, and a replacement for [theorem 3.1](#) is needed.

Research question 6 (Other digit distributions). Replace a uniform digit by a compactly supported distribution whose characteristic function has a controlled averaged algebraic logarithmic loss. Formulate necessary and sufficient hypotheses under which the cost is a scalar multiple of $\mathcal{Q}$. The exact convolution-power theorem gives one nontrivial family where the multiplier is known, but arbitrary characteristic zeros may change the conclusion.

Research question 7 (Local regularity and sharp boundary profiles). Locate the spatial points at which high mixed derivatives achieve their large norms. Determine whether the global non-Gevrey conclusion holds on every interior open set, and classify boundary directions where (9.2) has a matching lower bound. Global Fourier witnesses do not answer either spatial question.

Research question 8 (Effective constants and certified inversion). Replace the deliberately loose bad-index and sine-average constants with computable sharp or near-sharp ones. Combine them with interval sine products and validated multidimensional quadrature to make [theorem 9.3](#) a practical certificate generator for joint density values and derivatives.

Research question 9 (Formalization and broader envelope dualities). Formalize [theorem 5.1](#) first, then the analytic bridge to derivative norms. More generally, identify other random-series coefficient systems for which a finite-line or finite-cone envelope has an explicit positive-orthant conjugate. The present example shows that such a conjugate can encode a complete anisotropic regularity law rather than only a tail estimate.

13 Conclusion

The joint smoothness of common-digit Fabius laws is governed by an explicit finite matrix. The proof passes through three quantitative statements: a uniform upper bound that loses only finitely many cancellation indices; a matching lower envelope obtained by integrable sine-log averaging; and an exact convex conjugacy. Together they determine every mixed L^p derivative norm up to factors exponential in the total order.

The result both answers a precise version of the repository's sharp Fourier envelope question and adds information not present in a fixed-ray estimate. The signed resonance demonstrates why the hypotheses matter: smooth parameter dependence at every fixed order coexists with a discontinuous quadratic high-order profile. The remaining questions now concern secondary terms, local geometry, and transition regimes, rather than an unspecified leading growth constant.

A Explicit constants and a reusable proof checklist

The proofs give actual finite constants, although they are not optimized. A convenient collection is

$$B_0 = \sum_{i < j} \left(1 + \frac{2 \log(2d)}{|\lambda_i - \lambda_j|} \right), \quad (\text{A.1})$$

$$C_+ = \log 2 + B_0 + \frac{\log 2}{\lambda_{\min}}, \quad (\text{A.2})$$

$$C_d = (1 + 4\pi d) \log 2 + \log(2d), \quad (\text{A.3})$$

$$T_0 = \frac{1}{5(1 - e^{-2\lambda_{\min}})}, \quad (\text{A.4})$$

$$C_- = 1 + C_d \left(1 + \frac{1 + \log(2d)}{\lambda_{\min}} \right) + T_0. \quad (\text{A.5})$$

The last choice dominates the constant and the coefficient of S in (4.5). Put $c = 2 + C_+$ and

$$A_1 = 4^d \exp(C_+ + c^2 \mathcal{Q}(\mathbf{1})), \quad B_1 = \exp\left(c \max_j (M\mathbf{1})_j\right). \quad (\text{A.6})$$

Then (6.2) holds with these A_1, B_1 . For the derivative lower bound, (6.8) supplies

$$W^{-1} e^{-C_-} e^{-C_- \lambda_{\max} |\alpha|} e^{\mathcal{Q}(\alpha)}.$$

Taking maxima of these constants, and incorporating $(2\pi)^{-d}$ and W , gives the two constants in theorem 1.1.

A proof using this framework must keep six distinctions intact:

1. Signed parameters may be distinct while their moduli coincide; the bounded-transition argument needs distinct moduli.
2. A Fourier lower envelope is not a pointwise lower bound at every frequency; exact zeros cannot be discarded by assertion.
3. The logarithmic-box averaging variables are shared across factors; no independence of different factors is available or needed.
4. The conjugacy is on nonnegative derivative orders; the envelope is not one global quadratic polynomial.
5. The Fourier witness lower-bounds a norm on the entire support and does not locate a spatial maximizer.
6. The error constants are uniform in derivative orders and p , but not in vanishing modulus gaps.

B Package contents and rebuilding

The archive contains the TeX/PDF article, verification code, generated data, a JSON test report, provenance, requirements, and a README. The following commands run the computations and typeset the article:

```
python -m pip install -r requirements.txt
python code/verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

Run the last command three times to resolve references. The included data also allow PDF rebuilding without rerunning Python. No font files are redistributed; the README lists the required TeX packages.

References

- [1] Vladimir Reshetnikov / Provelt research corpus, *Common-Digit Fabius Zonoids: Exact volumes, hyperbolic-secant geometry, Bernoulli Gaussianization, and parameter jets*. Unrefereed repository report, inspected at commit 23dd71d2d3d5e85d2b97a8cc45f55e735d69a4ae. Relevant labels: thm:joint-char, thm:density-regularity, conj:ray-decay. [Pinned source in the Provelt repository](#).
- [2] J. Fabius, *A probabilistic example of a nowhere analytic C^∞ -function*, Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete 5 (1966), 173–174. Institutional preprint record (1965): [CWI publication 8160](#).
- [3] J. Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, arXiv:1702.05442 (2017), English translation of the author’s 1982 paper. In particular, Theorems 1–4 for the compactly supported function, its probability interpretation, and its explicit derivatives. [arXiv record](#).